



What is Communication?



Communication means exchanging information, ideas, messages, signs, voice, images or data from one person, place or device to another.

Face-to-face

Teacher explains in class

Voice

Phone call to parents

Text/Image

SMS, email, Messenger

Machine

Computer talks to
server



Why Is Communication Important?



Shares information quickly.



Connects people, offices, schools and banks.



Makes online classes, mobile banking and e-commerce possible.



Allows people and machines to exchange data.



Reduces distance and saves time.



Types of Communication



LOCAL COMMUNICATION



Occurs face-to-face or within a short distance.

EXAMPLES



Classroom discussion



Hand gesture



Two friends talking



KEY POINT

Direct and immediate communication.

REMOTE COMMUNICATION



Occurs over long distance using media/devices.

EXAMPLES



Phone call



Email



Video call



Radio / TV / Internet



KEY POINT

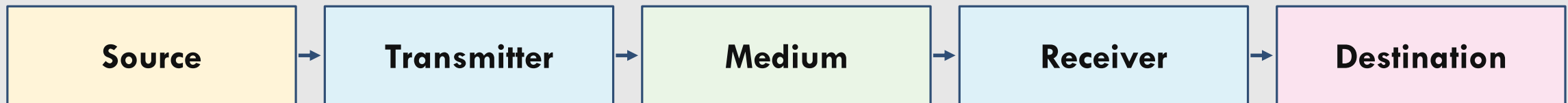
Uses devices and media to connect over long distances.



Concept of Communication System



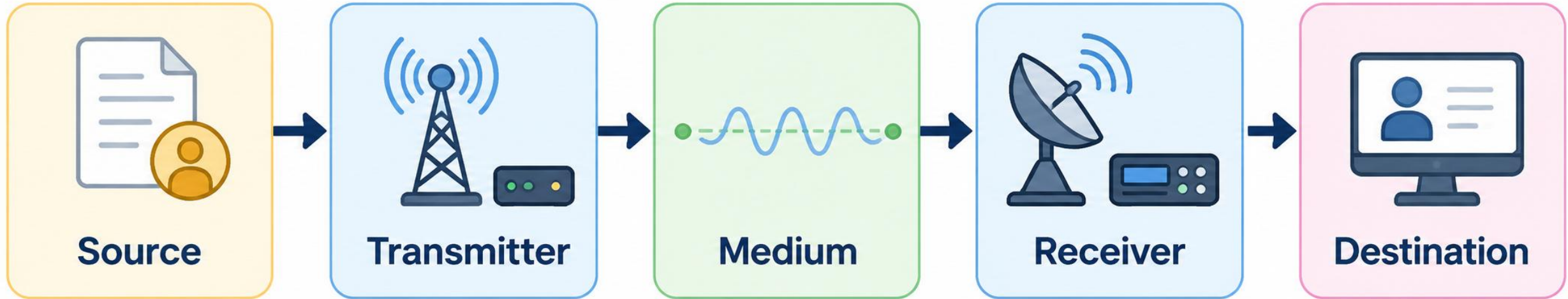
A communication system is a system used to transfer information from one place, person or device to another through a specific channel or medium.



Information must travel through a path and reach the correct destination.

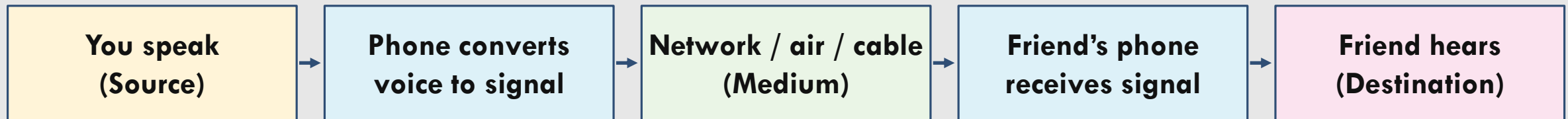


Concept of Communication System





Real-Life Example: Phone Call



Voice becomes signal → signal travels → signal becomes voice again

- ☐ If the signal is weak, the voice becomes unclear.
- ☐ If the receiver is off, communication fails.



Concept of Data Communication

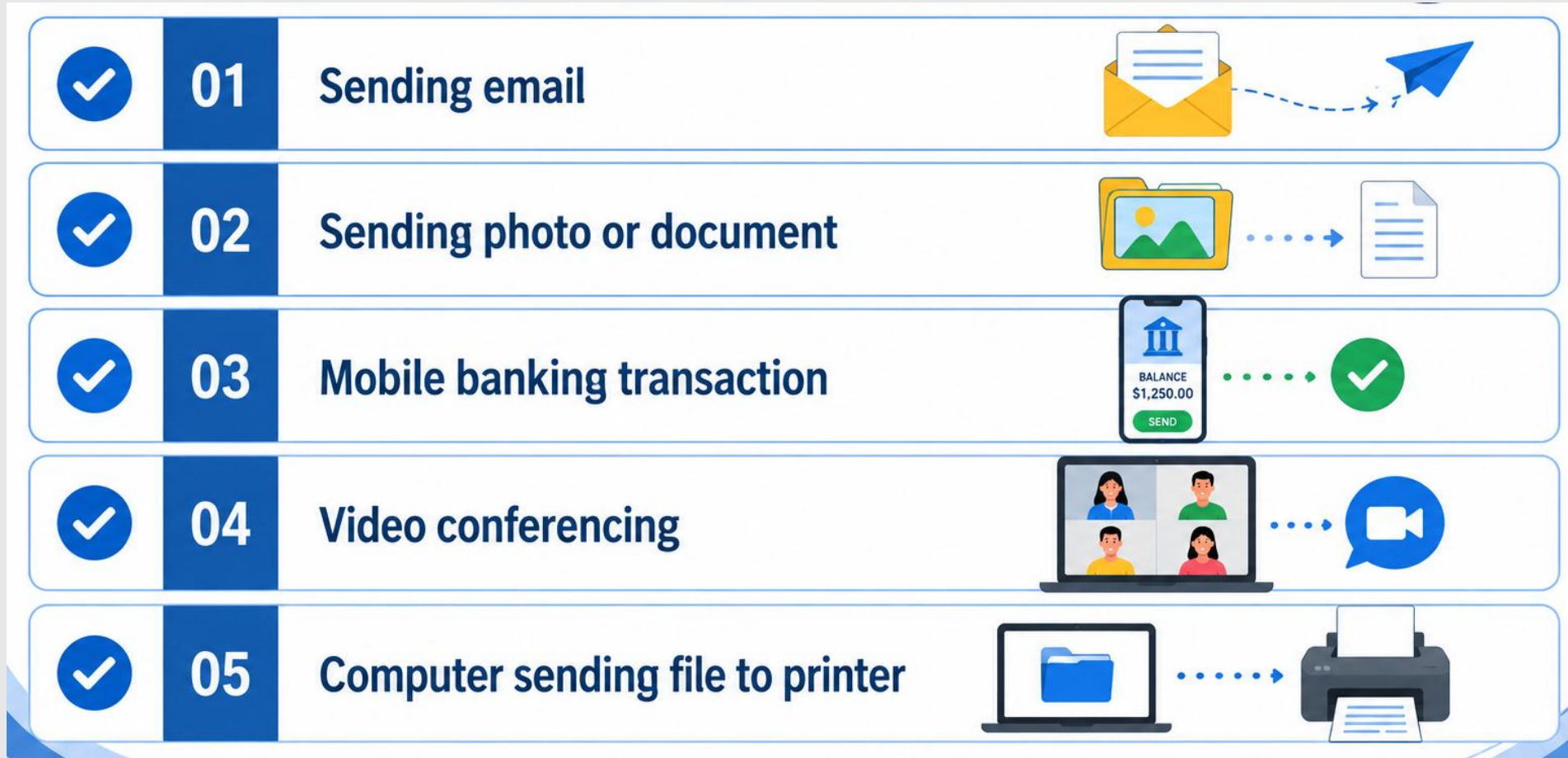


Data communication is the process of transferring data from one device, computer, person or place to another.

Data can be:
Text • Picture • Audio • Video •
Numbers



Concept of Data Communication





Why Is Data Communication Important?



Education

Google Classroom, online class, digital notes

Banking

ATM, card payment, bKash/Nagad, OTP

Office

Email, file sharing, printer, database

Daily Life

Messenger, YouTube, maps, video calls

Business

E-commerce, online order, delivery tracking

Without data communication, modern ICT services cannot work.



Characteristics of Data Communication



1. Delivery

Data must reach the correct receiver.

Example: Email goes to the right address.

2. Accuracy

Data must be correct and unchanged.

Example: Tk. 500 must not become Tk. 5000.

3. Timeliness

Data must arrive at the right time.

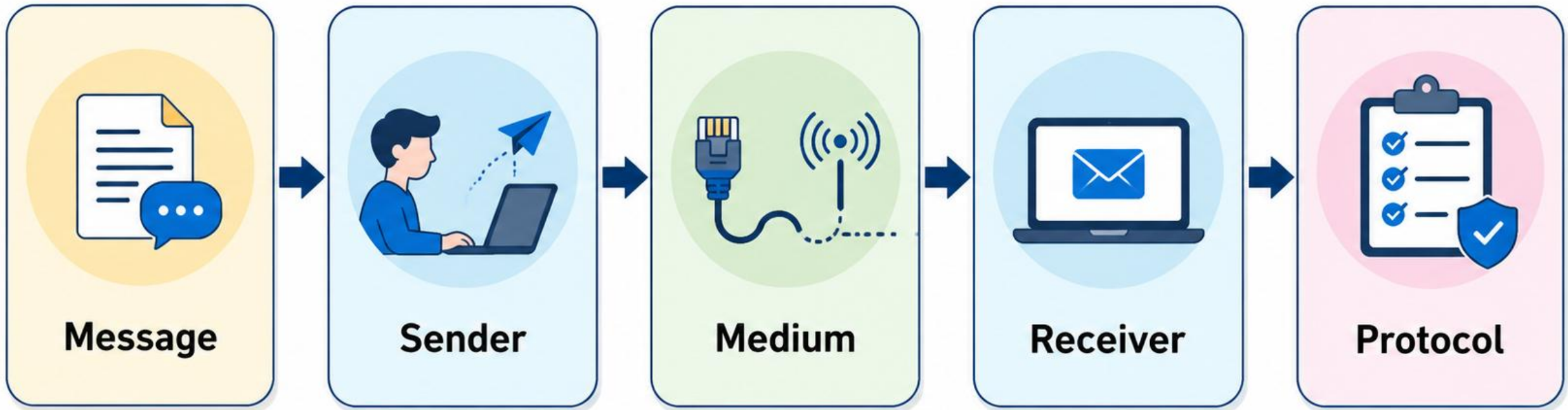
Example: Live video needs real-time delivery.

“Which characteristic fails if an OTP arrives after 10 minutes?”

“When you send a bKash/Nagad transaction, which is more important: delivery, accuracy, or timeliness?”



Basic Elements of Data Communication





Protocol: Rules for Communication



“Can two people communicate properly if one follows Bangla grammar and another follows only Japanese?”

Devices also need common rules. These rules are called protocols.

- ☐ HTTP helps web browsing.
- ☐ TCP/IP helps internet communication.
- ☐ Bluetooth rules help nearby devices exchange data.



Bandwidth



Bandwidth is the amount of data that can be transferred through a communication channel in a given time.

Bandwidth = Data transfer capacity per second

- ☐ Higher bandwidth → faster upload/download.
- ☐ Lower bandwidth → slower browsing and buffering.



Units of Bandwidth



Unit	Meaning
bps	bits per second
Kbps	kilobits per second
Mbps	megabits per second
Gbps	gigabits per second

- ☐ 1 Byte = 8 bits
- ☐ 1 KB = 1024 Bytes
- ☐ 1 MB = 1024 KB
- ☐ 1 kbps = 1000 bits per second
- ☐ 1 gbps = 1 Billion bits per second



ISP Connection: Shared vs Dedicated



Shared Connection

Bandwidth is shared among many users.

Example: home broadband in a locality.

Cheaper, but speed may drop during busy time.

Dedicated Connection

A fixed bandwidth is reserved for one customer/organization.

Example: bank, office, data center.

Stable and reliable, but costly.



Bandwidth Required by Various Services



Service	Approx. bandwidth	Service	Approx. bandwidth
Email	0.5 Mbps	Streaming movie	1.5 Mbps
Phone call / VoIP	0.5 Mbps	HD movie streaming	4.0 Mbps
Streaming music	0.5 Mbps	Video conferencing HD	4.0 Mbps
Web browsing	0.5–1.0 Mbps	Online HD gaming	4.0 Mbps
Video conferencing	1.0 Mbps	Internet game console	1.0 Mbps



Bandwidth Required by Various Services



If your home has 4 people watching HD video at the same time, is 5 Mbps enough?

Quick calculation: 4 HD streams need $4 \times 4 = 16$ Mbps



Bandwidth Math



Problem: If 20,000 bits are transferred in 5 seconds, find the bandwidth.

Formula: Bandwidth = Data transferred ÷ Time

$$\begin{aligned}\text{Bandwidth} &= 20000 \div 5 \\ &= 4000 \text{ bps}\end{aligned}$$

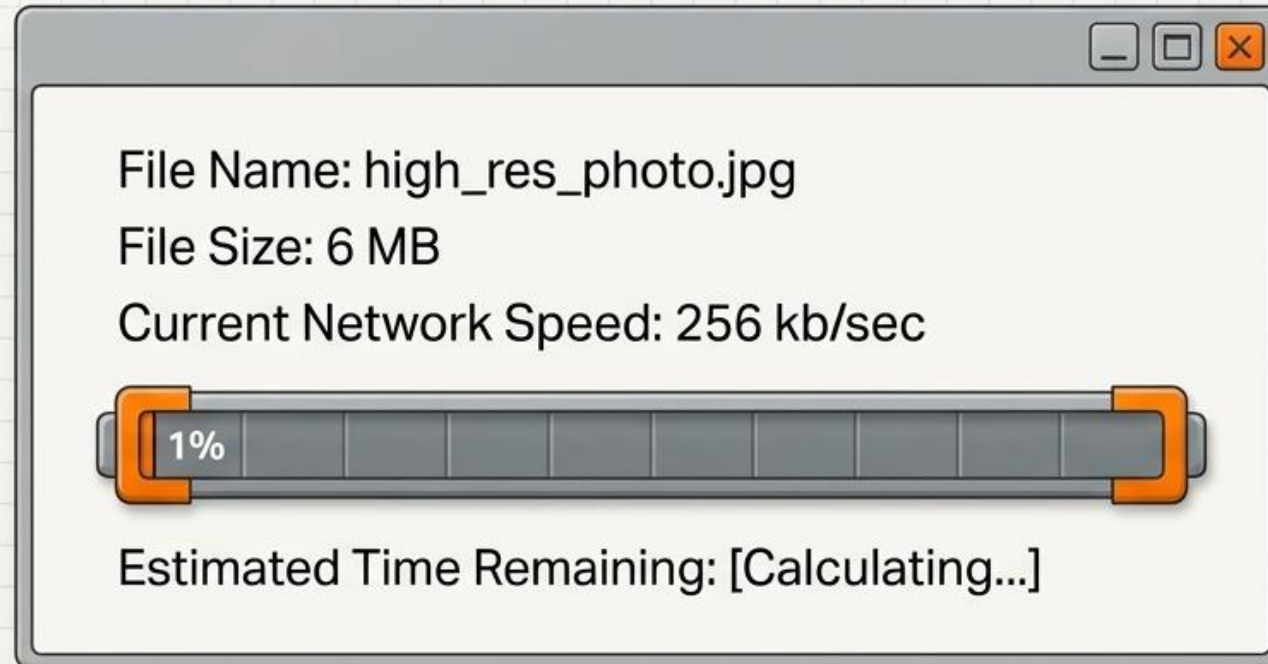
“If time decreases but data is same, what happens to bandwidth?”



Bandwidth Math



The 6MB Download



Grab a pen. How long will you actually be waiting for this download? (Hint: Watch your capital 'B's and lowercase 'b's!)



Bandwidth Math



Breaking Down the Download

Step 1: Convert Megabytes to Kilobytes

$$6 \text{ MB} = 6 \times 1024 \text{ KB} = 6144 \text{ KB}$$

6 MB



6144 KB

$\times 1024$

Step 2: Convert Kilobytes to Kilobits

$$6144 \text{ KB} \times 8 = 49152 \text{ Kb}$$

6144 KB



49152 Kb

$\times 8 \text{ bits/Byte}$

Crucial Step:
1 Byte = 8 bits!

Step 3: Divide data size by network speed

$$\text{Time} = 49152 \text{ Kb} \div 256 \text{ kb/sec} = 192 \text{ seconds}$$

49152 Kb

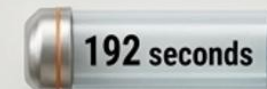


192 seconds

256 kb/sec

Step 4: Final Time

$$192 \text{ seconds} = 3 \text{ minutes } 12 \text{ seconds}$$





Data Transmission Speed



Data transmission speed can be divided into three types:

1. Narrow Band

Very slow speed
45–300 bps
Mainly text

2. Voice Band

Low to moderate speed
1 200–9 600 bps
Text + audio

3. Broadband

High speed
1 Mbps–1 Gbps
Text, image, audio, video



Data Transmission Speed



Narrow Band



Voice Band



Broad Band





Comparison: Narrow, Voice and Broadband



Subject	Narrow band	Voice band	Broadband
Speed	45–300 bps	1 200–9600 bps	1 Mbps–1 Gbps
Frequency	300–400 Hz	200–3600 Hz	1 MHz–300 GHz
Sent data	Text	Text, audio	Text, image, audio, video
Uses	Telegraphy	Telephone	Satellite, cable, DSL, fiber



Which One Will You Select?



Situation	Best choice	Reason
Sending only short text in old system	Narrow band	Text needs very low speed
Telephone line / simple audio	Voice band	Suitable for voice-frequency data
Online class, YouTube, gaming, file download	Broadband	Needs high speed and multimedia support
School computer lab	Broadband + dedicated if possible	Stable speed for many users

For modern internet use, broadband is the best option.